

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

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Date: 31-08-2026

1. TITLE

Design of a Student Profile Manager using JavaScript ES6 Classes and DOM Manipulation

2. AIM

To develop a web application that manages student profiles dynamically using JavaScript ES6 Object-Oriented Programming (OOP) classes, event handling, and Document Object Model (DOM) manipulation.

3. OBJECTIVE

The objectives of this experiment are:

- To implement Object-Oriented Programming principles in JavaScript using ES6 class syntax.
- To capture user input from HTML form input fields.
- To dynamically instantiate student objects with properties (Name, Roll, Department, CGPA).
- To render dynamic HTML content onto the webpage via DOM innerHTML properties.

4. THEORY

JavaScript ES6 introduced class syntax (class Student) providing a clear syntax for Object-Oriented Programming. Constructors encapsulate instance attributes (this.name, this.roll, this.department, this.cgpa). DOM (Document Object Model) manipulation via document.getElementById() allows reading element values and dynamically updating node tree content (innerHTML) in response to user interaction events (onclick).

5. ALGORITHM / PROCEDURE

1. Create index.html and define standard HTML5 document structure.
2. Design input fields for Name, Roll Number, Department, and CGPA with customized CSS styling.
3. Add a trigger button bound to the createProfile() JavaScript function via onclick attribute.
4. Create a target <div> container (id="profile") to hold generated student profile cards.
5. Define an ES6 Student class with a constructor accepting name, roll, department, and cgpa.
6. Write createProfile() function to extract input values using document.getElementById().value.
7. Instantiate a new Student object using the extracted data.
8. Format and inject HTML string into profile.innerHTML to display the formatted profile card.
9. Test and verify student profile creation in a browser.

6. CODE EXPLANATION

The HTML document includes styled text input fields and a button. CSS rules customize font family, margins, background color, input borders, and profile card shadow styling. Inside the <script> tag, the Student class defines a constructor setting instance properties (this.name, this.roll, this.department, this.cgpa). The createProfile() function reads values from input fields, instantiates a new Student object, and dynamically assigns HTML content to profile.innerHTML to display the formatted student profile card.

7. PROGRAM

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Profile Manager</title>
  <style>
    body {
      font-family: Arial;
      background-color: #356292;
      text-align: center;
      margin: 40px;
    }
    h1 {
      color: #d03636;
    }
    input {
      width: 380px;
      padding: 10px;
      margin: 6px;
      border: 1px solid #3eb475;
      border-radius: 5px;
    }
    button {
      background-color: #174a7e;
      color: white;
      padding: 10px 25px;
      border: none;
      border-radius: 5px;
      margin: 8px;
    }
    #profile {
      width: 430px;
      margin: 25px auto;
      padding: 20px;
      background-color: white;
      border: 2px solid #174a7e;
      border-radius: 8px;
      box-shadow: 3px 3px 8px #aaa;
      text-align: left;
    }
    #profile h2 {
      color: #174a7e;
      text-align: center;
    }
    #profile p {
      font-size: 16px;
    }
  </style>
</head>
<body>
  <h1>Student Profile Manager</h1>
  <h2>Enter Student Details</h2>
  <input type="text" id="name" placeholder="Enter Name">
  <br>
  <input type="text" id="roll" placeholder="Enter Roll Number">
  <br>
```

```

<input type="text" id="department" placeholder="Enter Department">
<br>
<input type="number" id="cgpa" placeholder="Enter CGPA">
<br>
<button onclick="createProfile()">Create Profile</button>
<div id="profile"></div>
<script>
    class Student {
        constructor(name, roll, department, cgpa) {
            this.name = name;
            this.roll = roll;
            this.department = department;
            this.cgpa = cgpa;
        }
    }
    function createProfile() {
        var name = document.getElementById("name").value;
        var roll = document.getElementById("roll").value;
        var department = document.getElementById("department").value;
        var cgpa = document.getElementById("cgpa").value;
        var student = new Student(name, roll, department, cgpa);
        var profile = document.getElementById("profile");
        profile.innerHTML =
            "<h2>Student Profile</h2>" +
            "<p><b>Name</b> : " + student.name + "</p>" +
            "<p><b>Roll No</b> : " + student.roll + "</p>" +
            "<p><b>Department</b> : " + student.department + "</p>" +
            "<p><b>CGPA</b> : " + student.cgpa + "</p>";
    }
</script>
</body>
</html>

```